Note

If $E_1, E_2, \cdots, E_k$ are elementary matrices (row operations) such that $(E_k \cdots E_2 E_1) A = I_n$, then $A^{-1} = E_k \cdots E_2 E_1$.

Transforming a non-singular matrix A to the form I_n by applying elementary row operations, is called **Gauss-Jordan method.** The steps in finding A^{-1} by Gauss-Jordan method are given below:

Step 1

Augment the identity matrix I_n on the right-side of A to get the matrix $[A \mid I_n]$.

Step 2

Obtain elementary matrices (row operations) $E_1, E_2, \cdots, E_k$ such that $(E_k \cdots E_2 E_1) A = I_n$.

Apply $E_1, E_2, \cdots, E_k$ on $[A \mid I_n]$. Then $[(E_k \cdots E_2 E_1) A \mid (E_k \cdots E_2 E_1) I_n]$. That is, $[I_n \mid A^{-1}]$.

Example 1.20

Find the inverse of the non-singular matrix $A = \begin{bmatrix} 0 & 5 \\ -1 & 6 \end{bmatrix}$, by Gauss-Jordan method.

Solution

Applying Gauss-Jordan method, we get

$$[A \mid I_2] = \left[\begin{array}{cc|cc} 0 & 5 & 1 & 0 \\ -1 & 6 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} -1 & 6 & 0 & 1 \\ 0 & 5 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \to (-1)R_1} \left[\begin{array}{cc|cc} 1 & -6 & 0 & -1 \\ 0 & 5 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to \frac{1}{5}R_2} \left[\begin{array}{cc|cc} 1 & -6 & 0 & -1 \\ 0 & 1 & (1/5) & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 + 6R_2} \left[\begin{array}{cc|cc} 1 & 0 & (6/5) & -1 \\ 0 & 1 & (1/5) & 0 \end{array}\right].$$

So, we get $A^{-1} = \begin{bmatrix} (6/5) & -1 \\ (1/5) & 0 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 6 & -5 \\ 1 & 0 \end{bmatrix}$.

Example 1.21

Find the inverse of $A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ by Gauss-Jordan method.

Solution

Applying Gauss-Jordan method, we get

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccc|ccc} 1 & (1/2) & (1/2) & (1/2) & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 3R_1} \left[\begin{array}{ccc|ccc} 1 & (1/2) & (1/2) & (1/2) & 0 & 0 \\ 0 & (1/2) & -(1/2) & -(3/2) & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to 2R_2} \left[\begin{array}{ccc|ccc} 1 & (1/2) & (1/2) & (1/2) & 0 & 0 \\ 0 & 1 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

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